

Operator–Operand Factorization in LLM Residual Streams: Causal Influence and Compositional Sufficiency

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Abstract

How do language models represent a relational operation — *currency-of*, *capital-of* — as distinct from the entity it applies to? We separate three levels of evidence. **Representation:** a two-way effects decomposition of the query-position state is ~90% additive in operand and operator across Qwen3-1.7B/8B and Gemma-2-9B, and the steering vector is *identically* the decomposition's operator main effect. **Causal influence:** adding the operator difference flips the target-vs-source answer margin for every ordered relation pair in all three models; the decisive nulls — directions rebuilt under **permuted relation labels**, and random directions **inside the operator subspace** — abolish the effect. **Behavioral sufficiency:** a per-layer residual trajectory *composed* from the decomposition's parts, $\mu + \text{operand} + \text{operator}$, patched at the query position, makes the model *produce* the target answer at its own accuracy level; components built without the target cell still generate, and swapping the operand component redirects the answer. The interaction term adds nothing detectable — and the earlier failure of additive steering to generate was a **dose artifact**: an out-of-sample-calibrated dose generates at the same level, while overdosing inflates the margin as it pushes the state off-manifold. The structure generalizes (held-out operands, re-worded prompts, a second domain), separates the operation from its surface word, and does not emerge for arithmetic or logic under the identical pipeline.

1 Introduction

When a language model completes "*The currency of Italy is—*", does it represent *currency-of-Italy* as one fused thing, or as an entity plus an operation applied to it? This paper argues, causally, for the second reading — and asks the question a claim like that must answer at **three separate levels of evidence**: **representation** (does a sta-

ble relational direction *exist*, separable from the operand?), **causal influence** (does intervening on it systematically move the model's *relative preference* between answers?), and **behavioral sufficiency** (can the factorized parts make the model actually *produce* the target answer?). The levels matter because margin-moving interventions are routinely read as if they controlled behavior. Here that gap becomes the object of study — and resolves: it is a property of *how* the direction is applied (mode and dose), not of *what* it contains.

A line of work shows that a *task* or *relation* can be captured by a single addable vector in an LLM's residual stream (task vectors, Hendel et al., 2023; function vectors, Todd et al., 2024), that many relations are approximately linear operators (LRE, Hernandez et al., 2024; Merullo et al., 2024), and that recall can be read as function execution over components at different positions (Wang and Xu, 2025). What has not been quantified is the **joint structure of operator and operand within one prediction state**: whether the operation *factorizes* from its argument there, *generalizes*, is separable from the *word it produces*, and suffices to *drive generation* — and whether any of this holds beyond relational facts. We answer these on Qwen3 and Gemma-2, across a geographic and an animal-taxonomy domain, with arithmetic and logic as the contrast. A closing discussion (§5) offers **declension** as the organizing metaphor for the asymmetries observed.

2 Related work and what is new here

Extracting relational concepts, and knowledge recall as function execution. The closest prior line is Wang et al. (2024), who locate hidden states expressing a relation separately from its subject and transplant them between subjects — and its successor Wang and Xu (2025), which abstracts recall into a functional structure (subject activa-

tions as *input argument*, relation activations as *function body*, object activations as *return value*), verified by patching and counter-knowledge swaps of each component. Our object of study differs in four ways: they localize and exchange components read at *different token positions*, while we quantify the **joint operator $\times$ operand decomposition of a single prediction state**; we show the steering vector *is* that decomposition's operator main effect, numerically; we validate against **competitive semantic nulls** (permuted labels, within-subspace random); and we establish **generative sufficiency by additive reconstruction**, including components built without the target cell and an out-of-sample dose calibration. Their framing is function *execution across positions*; ours is state *composition at one position* — complementary readings of the same recall machinery.

Operation as an addable vector; relation as a linear operator. Task Vectors (Hendel et al., 2023) and Function Vectors (Todd et al., ICLR 2024) extract one vector per ICL task and add/patch it (similarly Liu et al., 2024; Turner et al., 2023) — but bundle operator *and* operand into one unfactored direction, without competitive nulls. LRE (Hernandez et al., 2024) fits a *monolithic* affine map per relation. Summing Up the Facts (Chughtai et al., 2024) decomposes recall into additive *circuits*; ours is a two-way effects decomposition of the **residual state itself**, with a measured interaction term and a sufficiency test on the components.

Attribute subspaces / reading position. RAVEL (Huang et al., 2024) disentangles multiple *attributes of one entity* into subspaces (operators sharing an operand); we measure the orthogonal axis, **operator vs operand**. The operand $\rightarrow$ operator shift along the sequence restates the subject-enrichment $\rightarrow$ attribute-extraction dynamics of Geva et al. (2023) in operator/operand terms — our least novel point, included as confirmation.

Global workspace and the Jacobian lens. Gurnee et al. (2026) introduce the Jacobian lens and identify a mid-depth "global workspace" band. We adopt their depth band as the site of intervention; our contribution is orthogonal — the operator/operand structure of what the band *holds* — and we report a controlled null on the lens's readout advantage for our task (§4.9).

Arithmetic and logic. LLM arithmetic is reported as a "bag of heuristics" (Nikankin et al.,

2025) and Fourier/helical procedures (Nanda et al. 2023; Zhong et al., 2023; Kantamneni and Tegmark, 2025), with the operand *value* cleanly linear (Gurnee and Tegmark, 2024); truth is steerable (Marks and Tegmark, 2023) and logical-operator heads exist (Hong et al. 2024). No prior work reports a **cross-domain operator/operand factorization**; we provide one and show where it does not emerge.

What is new here is the combination, not any single ingredient. That relations admit addable vectors and linear decoders, and that recall components can be located and swapped across positions, is established; what has not been done is the joint, quantified analysis at one position: the operator $\oplus$ operand **factorization of the residual state itself**, with the steering vector shown to *be* the ANOVA operator component; the all-pairs swap validated against **permuted-label and within-subspace nulls; held-out-operand generalization**; the **compositional sufficiency** test (does a state assembled from the parts *generate*?) with its dose $\times$ position calibration; the **operation-vs-realization dissociation**; and the **cross-domain test** (each claim, its evidence, and its decisive control are tabulated at the head of §4). The declension reading is our *presentation* of these measurements, not an additional empirical claim.

3 Method

Setup. Qwen3-1.7B/8B throughout, plus Gemma-2-9B for the cross-architecture replication (§4.7); all run in bf16 on a single AMD Strix Halo APU (environment details in the supplementary repository). A relation is rendered into a template; the canonical frame is "The {op} of {a} is", and §4.5 additionally uses two paraphrase frames (question–answer and discourse-prefixed) that hold the {op} of {a} unit fixed while varying the surrounding frame, all ending in *is* so answer scoring is comparable. Reading position is the query token unless stated; the depth "workspace" band (38–92% of layers) is where Gurnee et al. (2026) locates the global workspace and where we build operator directions.

Notation. The model's state is a grid $h_{\{\ell, t\}} \in \mathbb{R}^d$: one residual-stream vector per layer ℓ per token position t ; every measurement fixes both indices. Two reading positions recur: the **entity token** (the operand's last subword, $t = -2$ in the canonical frame) and the **query token** (the fi-

nal prompt token, $t = -1$, where the next-token prediction is read; "query" means reading position, not attention's Q). Rendering the operator-operand grid through one template gives, at each (ℓ, t) , a family of states $h_{\{\ell, t\}}(o, k)$.

Operator direction $\equiv$ factorization component. For operator k , $v_{\ell}(k) = \text{mean}_o[h_{\{\ell, -1\}}(o, k) - \text{mean}_{k'}h_{\{\ell, -1\}}(o, k')] - \text{one direction per layer}$ of the workspace band (13 layers per model), the function-vector construction. This is *algebraically identical* to the operator main effect of the two-way factorization below ($b_{\ell}(k) = \text{mean}_o h_{\{\ell, -1\}}(\cdot, k) - \mu_{\ell}$), and we verify the identity numerically (max relative deviation 8.5×10^{-8} across the band). Everything the paper does — steering, composing, decoding — manipulates this one object; the differences between experiments are differences in *how it is applied*, never in what is extracted. We characterize it as a **relation-conditioned retrieval operator** — it conditions retrieval on the queried relation; we make no claim about the attention/MLP circuitry that implements the binding.

Two-way effects decomposition. At a fixed (ℓ, t) : $h = \mu + a(o) + b(k) + e(o, k)$ — operand main effect, operator main effect, and **interaction (cell residual)**, every term carrying the (ℓ, t) indices. The identity is exact because the design is balanced and e is defined as the residual; this is a descriptive decomposition, not an inferential ANOVA, and with one observation per cell e also absorbs any cell-specific idiosyncrasy. Variance shares are $SS_a = K \cdot \sum_o \|a(o)\|^2$, $SS_b = O \cdot \sum_k \|b(k)\|^2$, $SS_e = \sum_{\{o, k\}} \|e(o, k)\|^2$, each reported as its fraction of $SS_a + SS_b + SS_e$ (O operands, K operators). Table 1 uses the mid-workspace layer — **layer 17 of 28 (63% depth) at 1.7B, 23 of 36 at 8B, 27 of 42 on Gemma-2-9B** — at $t = -1$ and, as the control, $t = -2$.

Swap, positions, and controls. Efficacy of $k_A \rightarrow k_B$ is the change in $\text{logit}(\text{answer}_B) - \text{logit}(\text{answer}_A)$ at the query position when $\alpha \cdot (v(k_B) - v(k_A))$ is added — over the band or a single layer, at all prompt positions or a chosen one (all / query / operand / sentence-initial control). Beyond the **matched-norm random** direction, the null battery includes directions rebuilt under

permuted relation labels (per-operand and global — same statistics, no semantics: the decisive null), random directions **inside the operator subspace**, the correct direction at the **wrong layer** and **wrong position**, an **other-relation** direction of matched norm, and **shuffled-answer** scoring. Every intervention reports margin change, normalized margin, target rank before/after, top-1, short-decode containment (§below), on-task KL, and off-task KL on unrelated WikiText — not flips alone. Answers are scored at their distinguishing first token (bare digit for arithmetic); a tokenization guard drops, per operator pair, operands whose answers collide there (4 of 12 survive on the syncretic demonym $\leftrightarrow$ language pairs; all 12 elsewhere).

Composition (behavioral sufficiency). The factorization's component vectors, computed per layer, are recombined and **replace** the residual at the query position ($\mu + \text{operator}, \mu + \text{operand}, \mu + \text{operand} + \text{operator}, \mu + \text{interaction}$, the full donor, a norm-matched magnitude control) — plus two loophole-closers: a **leave-one-cell-out** reconstruction whose components never saw the target cell, and a **wrong-operand** composition that should (and does) redirect the answer. Three named generation readouts, used consistently and never cross-compared: **short-decode containment** (3-token greedy continuation contains the gold answer, case-insensitive — *the euro, euros, Euro* all count; the model's clean accuracy under the same readout is the ceiling); **long-decode classification** ($k = 8$, each raw generation labeled target / source / wrong-entity / wrong-relation / degraded); and **forced-choice sequence probability** (length-normalized log-probability of each gold answer under the same intervention — immune to article interception and multi-token variants).

Generalization and domains. Build $v(\text{op})$ from half the operands; test the swap on the held-out half — a genuine operator vs. interpolation among build examples. Domains: five country relations (currency, capital, language, demonym, continent $\times$ 12 countries), an independently curated **animal-taxonomy domain** (class, habitat, diet, covering $\times$ 12 animals; stable 1-to-1 answers, tokenizer-screened, three paraphrase frames), and arithmetic / comparison logic as the contrast domains.

Statistical treatment. The ordered swaps are **not** independent observations: they share opera-

tor directions and operands. We report cluster-bootstrap percentile intervals (10,000 replicates) at two levels: within a pair the **operand** is the resampling unit; across the paradigm the **operator** is the top-level cluster (a dyadic node bootstrap — resample operators with replacement, weight each pair by its endpoints' multiplicities, resample operands within surviving pairs). Because only five operator clusters exist, we additionally report **leave-one-operator-out**: dropping any operator leaves the contrast in [+18.8, +25.8] at 1.7B and [+23.3, +28.9] at 8B with flip fraction 1.00 throughout — no single operator carries the paradigm. Composition claims (§4.2) use paired per-cell differences under the same hierarchical bootstrap. Per-operand values are released as long-form artifacts (Appendix A).

J-space controls. We repeat readout-geometry in the J-lens readout `unembed(J·h)` vs the logit-lens readout `unembed(h)`, and re-run efficacy with **spectrum-matched random-projection** and **permuted-vocabulary** null lenses (Appendix B).

4 Results

Each experiment below tests one claim against the control that could kill it:

4.1 Relational operators are causally manipulable — and the effect is localized and label-specific

All 20 ordered operator swaps flip the **target-vs-source logit margin** while the matched-norm random control is ≈ 0 . Under the operator-level cluster bootstrap (§3), the swap-random contrast is **+22.6, 95% CI [+14.0, +32.1]** at 1.7B, **+26.0 [+17.9, +32.8]** at 8B, and **+30.4 [+23.8, +34.8]** on Gemma-2-9B; the flip fraction is **1.00 [1.00, 1.00]** in all three. Per-pair operand-bootstrap CIs never cross zero, and the weakest pairs are exactly the syncretic ones (demonym $\leftrightarrow$ language, which share their surface form; per-pair distributions in Appendix A).

Where the vector has to land (1.7B, $\alpha = 4$). Injecting at the **query token alone** reproduces the all-position effect on the margin (+28.7 [+16.1, +39.8] vs. +28.9 [+16.2, +39.9]) at **60× lower off-task cost** (KL 0.29 vs. 18.4 nats/token), and one late layer at the query token brings the target to median rank 69 at near-zero cost (0.04); the operand token and a structurally irrelevant token

do essentially nothing (+1.8 / +2.3). Full position $\times$ scope battery in Appendix A.

The margin effect is a **localized edit of the queried relation** — position-specific (sign-correct 98/98/91% for the effective conditions vs. 4–16% for the controls) — not a global perturbation; it replicates at 8B (query-only +32.7 vs. all +33.2, at $63\times$ lower off-task cost).

The label is everything. Directions rebuilt from the *same* residuals with **permuted relation labels** — same statistics, no semantics — abolish the effect: **+0.7 [-0.8, +3.9]** per-operand, +0.6 global (20 redraws; 8B +0.15 / +0.09 vs. real +26.0). A random direction **inside the operator subspace** also does nothing: +0.75 [-2.2, +3.5] (8B +0.43). Three probes stay nonzero for *mechanistically informative* reasons and decompose the margin: an **other-relation** direction keeps the shared $-v(\text{source})$ and yields $\approx$ half the effect (+11.1 — uninstalling the source is half the margin); **shuffled-answer scoring** still moves (+21.1) because $+v(\text{target})$ boosts the whole answer *category* — operand specificity lives in generation, which §4.2 measures; and the **wrong (early) layer** retains +9.8 because residual additions *persist* downstream (locality measured properly in §4.6; forest plot in Appendix A).

Dose-response, both signs. The effect is a **signed causal axis**: the operator-specific shift is monotone in α through zero and *reverses* with the direction — +22.6 at $\alpha = +4$ vs. **-21.0 at $\alpha = -4$** , an almost exact mirror (8B: +26.0 vs. -24.5), saturating at both ends (the positive branch higher, +29 vs. -15 raw, because the clean margin already favors the source: a floor) — while the random control's shift ($\approx +6$) is flat in $|\alpha|$, which is why every headline number is a swap - random contrast. Off-task cost grows with dose and breadth (7.9 $\rightarrow$ 21 nats/token band-wide vs. 0.29 at the query token). What dose does to *generation* is the subject of §4.2. (Curves in Appendix A.)

4.2 A composed residual trajectory generates the answer

At the default dose, additive steering flips every *margin* but rarely makes the model *say* the target answer (0.4–2.2% short-decode containment), whereas replacing the query-position residual with a **real donor activation** (same operand, target relation) does — 51%, at the model's clean accuracy of 53%. Is the donor carrying something the direction lacks — operand binding, a non-additive

claim	main evidence	decisive control	scope
a stable operator direction exists	ANOVA ~90% additive; $v(\text{op}) \equiv$ main effect (§4.3)	reading-position control	9 relations, 3 models
it causally controls the margin	all-pairs swaps flip, 3 models (§4.1)	permuted labels $\rightarrow 0$; subspace-random $\rightarrow 0$	relative preference
the effect is localized	query token $\approx$ all positions, $60\times$ lower KL (§4.1)	operand / sentence-initial positions ≈ 0	both Qwen3 scales
the operator generalizes	held-out operands; re-framed, re-worded prompts (§4.5)	random ≈ 0 ; transfer $\approx$ within	relations + animals
the parts suffice to generate	composed patch $\approx$ donor $\approx$ ceiling; calibrated steering (§4.2)	leave-one-out; wrong-operand; magnitude; interaction	greedy, this setup
the factorization is domain-scoped	animals replicates; arithmetic/logic do not (§4.8)	held-out fails off-relation	our setup (§4.8)

Table 1: The paper’s claims, each with its main evidence, the control that could kill it, and its scope.

interaction, lexical or execution signals? We decompose the donor by the exact factorization and patch partial reconstructions back in. To be precise about the object: what is patched is a **per-layer factorized residual trajectory** — one reconstructed vector per workspace-band layer, all at the query position — not a single-layer vector (the single-layer variant is reported at the end of this section; every variant includes μ , since the hook *replaces* the state and a replacement must be a plausible full-magnitude residual). 1.7B, 224 (pair, operand) cells:

The donor carries nothing detectable that the additive parts don't. The paired per-cell difference composed - donor is **+0.9 pp, 95% CI [-6.9, +8.3]** at 1.7B (51.8% vs. 50.9%; clean ceiling 53%) and -3.1 pp [-12.5, +5.2] at 8B (62.1% vs. 65.2%; ceiling 68%); adding the interaction term changes nothing detectable (-0.9 pp [-8.3, +6.9]; interaction *alone*: 7.1%, near the 4.5% magnitude control). Honest limit: with five operator clusters a formal ± 5 pp equivalence test is underpowered — the claim is "no detectable gap," not certified equivalence. Two controls close the loopholes. A **leave-one-cell-out** reconstruction, built *without ever seeing the target cell*, still generates at 35.7% (8B: 50.4%) — composition, not leakage — though genuinely below the donor (paired -15 pp, CI excluding zero at 8B). And a **wrong operand's** component redirects the answer to *that* operand (34.4% says-other vs. 1.8% baseline; 8B: 48.2%); the operator selects the relation, the operand selects whose answer.

Sufficiency by steering, at the calibrated dose. Since $v(\text{op})$ is the operator component (§3), adding $\alpha \cdot (v(B) - v(A))$ at the query position of one layer lands, at $\alpha = 1$, on $\mu + \text{operand} + \text{operator}(B) +$

$\text{interaction}(A)$ — the composed state up to a wrong interaction term. The dose sweep confirms it behaviorally: generation follows an **inverted-U peaked exactly at the on-manifold dose** — 38.4% at $\alpha = 1$ for a single layer, **51.3% at $\alpha \approx 0.1$ for the 13-layer band** (band additions accumulate, so the band's on-manifold dose is $\approx 1/13$ per layer; the same peak doses reproduce at 8B: 45.1% and 56.2%) — while past the peak the margin keeps climbing (+28.9 at $\alpha = 4$) as generation collapses. The dose is not tuned post hoc: **calibrating α on half the operands and evaluating frozen on the disjoint half**, α^* is identical in all five partitions (0.1 band, 1.0 single layer), held-out generation at frozen α^* is 60.8% [56.2, 67.3] (single layer 42.2%), and the evaluation-optimal dose coincides with the calibrated one in 10/10 splits. The earlier "steering moves margins but cannot generate" result was an **overdose artifact**, not an information deficit: the factorized representation is **compositionally sufficient**, by replacement and by calibrated addition. (Single layer: $\mu + \text{operator}$ alone generates at 44.6% — unpatched layers still carry the operand.)

The generation audit (Appendix A) closes the metric loopholes directly. Classifying every raw $k = 8$ generation: the calibrated dose produces the *fluent, correct* answer (59.4% target vs. 0.9% source; composed patch 57.6% — both at the clean prompt's own 58.5%), while at the overdose **94.6% of generations are degraded token loops** by manual inspection — category-correct, fluency-destroyed, not surface variants the metric missed. A **forced-choice** readout (full-sequence log-prob among the operand's gold answers, under the intervention) then dissociates information from fluency: the target wins at **80.4% even at the overdose** (87.5% calibrated, 84.4% donor patch,

patched state	says target	says source	says other	rank(target), median	top-1
magnitude control (μ + norm-matched random)	4.5%	33.5%	0.0%	696	0%
interaction only (μ + inter)	7.1%	35.3%	0.0%	166	0%
operand only (μ + operand)	7.6%	37.5%	0.0%	54	0%
operator only (μ + operator)	20.1%	14.7%	1.8%	71	0%
operator + operand (μ + both)	51.8%	13.4%	1.8%	2	31.7%
operator + operand, held-out cell	35.7%	18.8%	1.8%	5	17.9%
operator + wrong operand	20.1%	11.6%	34.4%	206	0%
full donor (μ + both + inter)	50.9%	12.5%	1.8%	2	31.2%

Table 2: Decomposing the donor activation (1.7B, 224 cells; clean greedy ceiling 53%): a state composed from the factorization’s additive parts makes the model emit the target answer as well as the real donor does; the interaction term is not needed. "Held-out cell" rebuilds the components without ever seeing the target cell; "wrong operand" redirects the answer to the swapped operand.

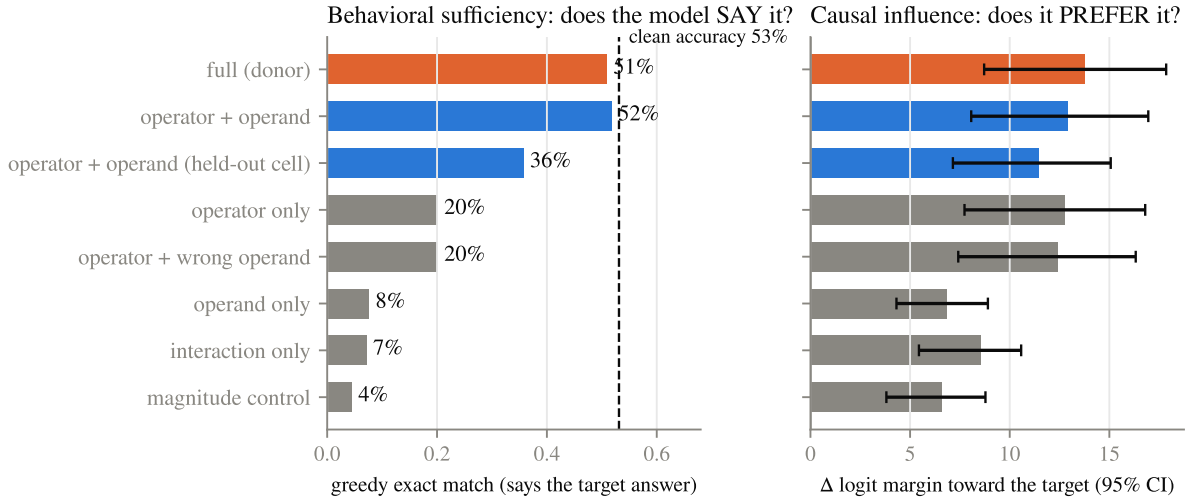


Figure 1: Decomposition ladder: short-decode containment and margin change for each partial reconstruction of the donor activation. The additive composition matches the full donor at the model’s clean ceiling; interaction-only and magnitude controls do nothing; the held-out-cell reconstruction and the wrong-operand redirect close the leakage and specificity loopholes.

2.7% clean). Overdosing destroys the generation manifold, not the encoded preference.

4.3 The representation factorizes into operator $\oplus$ operand

Does the state literally decompose into "the thing" plus "the question asked about it"? Two-way ANOVA of $h_{\{\ell, t\}}(o, k)$ at the mid-workspace layer (L17/28 at 1.7B, L23/36 at 8B; §3) — one decomposition per reading position:

~90% additive; operand and operator components occupy distinct subspaces (principal angles 41–85° at the query token — near-orthogonality is the high-dimensional default, so the load-bearing evidence is the variance split and the causal results). Since the steering vector is numerically this decomposition’s operator main effect (§3), the direction that flips margins (§4.1), the component that composes into a generating state (§4.2), and

the term that dominates this split are **one vector seen three ways**. The entity enters operand-dominant and is re-marked by its queried role at the query position (the reading-position control rules out template echo).

(A PCA of the 60 workspace states, showing the operand-organized cloud at the entity token reorganizing into operator clusters at the query token, is in Appendix A.)

4.4 Operation $\neq$ realization (syncretism)

language-of and *demonym-of* emit the **same word** (Italian) yet are **distinct operator directions**; the swap between them is weak precisely because they share their output word. An **exponent-free operator marker** — "exponent" in the morphological sense: the surface form that realizes a feature, here the shared answer token — built from the shared-output pairs

read position	operand	operator	interaction (cell residual)
query token	5% / 6%	86% / 82%	9% / 13%
entity token	59% / 55%	31% / 34%	9% / 11%

Table 3: Two-way ANOVA of the workspace state: variance shares by read position (1.7B / 8B).

($\text{mean}[h(\text{language}) - h(\text{demonym})]$, the output word cancelled by construction) still installs the relation causally (1.7B: $-3.6 \rightarrow +12.5$): the operator representation is not an output-word representation (§5 reads this as syncretism).

(Operator-direction cosines and the exponent-free-marker bar chart are in Appendix A.)

4.5 Operators generalize: held-out operands, prompt frames, wordings — and a second domain

Held-out operands. $v(\text{op})$ built on 6 operands and applied to the 6 held-out still flips **20/20** swaps — a transferable operator, not interpolation: contrast **+20.0** [**+10.8**, **+29.5**] at 1.7B, **+22.8** [**+14.0**, **+31.0**] at 8B, flip fraction 1.00 at both. This is not one lucky split: over five shuffled partitions every pair flips in every partition (contrast range [+20.0, +22.9]; 8B [+22.8, +25.6]), and full leave-one-country-out gives 220/224 flips at 1.7B and 222/224 at 8B.

Frames and wordings. Across three paraphrase frames, every build→test combination flips — 1.7B **180/180**, *frame-invariant to two decimals* (+22.6 on all test frames); 8B 100/100. Stronger: the direction survives re-lexicalizing the *relation itself* (*currency of* → *money used in*, ...) both ways, **40/40 at both scales**, transfer $\approx$ within (+22.59 vs. +22.62). Clean baselines shift; the causal effect does not: it is a property of the **relation**, not its wording.

A second, non-geographic domain. Every signature replicates on an independently curated animal-taxonomy domain (class / habitat / diet / covering $\times$ 12 animals; 1.7B): **12/12** ordered swaps flip (+18.1 [+13.5, +22.8]); held-out operands **12/12** at +18.0 — transfer indistinguishable from within; cross-frame **72/72**, again frame-invariant to two decimals; permuted-label and subspace nulls at zero (-0.5 and +0.2, CIs spanning 0); the same reading-position signature (operand 79% at the entity token, operator 62% at the query token); and even the same quantitative margin decomposition (the other-relation probe retains $\approx$ half the effect in both domains: 47% vs. 49%).

Two honest caveats: the animal factorization carries more interaction (22% vs. 9% — informative in §4.8), and the generation-level readout is floor-limited here (the model's own clean greedy accuracy on these multi-token answers is 6%), so in this domain sufficiency is attested by margins and rank, not exact match.

4.6 Where the operator is represented is not where it acts

A per-layer sweep (1.7B; direction built **and** injected at each single layer) separates three notions the band framing conflates. *Trivial readability*: a held-out-operand classifier decodes the operator at **100% from every layer including layer 0** — the operator's token is in the prompt; decodability locates nothing. *Representation*: the operator share peaks **early** (92% at 19% depth), decaying to 57%. *Causal leverage*: ≈ 0 early, peaking **late** (+21.3 at 89% depth $\approx$ the full band's +22.6). The informative curves peak **70 depth-points apart** (8B: 23% vs. 89%): where the operator is most cleanly factorized is not where injecting it most moves the output — being *readable*, *the representation*, and *causally potent* are three different layer profiles of one direction (explains the wrong-layer probe of §4.1; figure in Appendix A).

4.7 Cross-architecture replication (Gemma-2-9B)

The entire pipeline transfers unchanged to **Gemma-2-9B** — different corpus, tokenizer, and architecture family. Every qualitative signature replicates, with the strongest effects of the three models: all-pairs **+30.4** [**+23.8**, **+34.8**] (20/20), held-out **+26.6** [**+20.6**, **+33.2**], operator variance 84.8% with 6.8% interaction (full table in Appendix A). Raw logit units are not comparable across architectures (Gemma-2 soft-caps logits) — compare flip rates and variance shares, which match or exceed Qwen3's: the factorization is not an idiosyncrasy of one family.

4.8 Domain scope: a second retrieval domain replicates; arithmetic and logic do not emerge

If the factorization were an artifact of prompting, it should appear wherever the prompt has the same shape. It does not. Extending the identical pipeline to arithmetic (+, ×, -) and comparison-logic operators, on 1.7B, with the animal domain (§4.5) as the second retrieval column:

domain	op var	interaction	held-out	swap/rand
countries	86%	9%	20/20	+21 / ~0
animals	62%	22%	12/12	+19 / +0.7
arithmetic	55%	23%	2/6 (≤rand)	+0.4 / +0.3
logic	33%	34%	2/6 (≤rand)	+1.2 / +0.2

The gradient replicates and sharpens at **8B** (arithmetic held-out 1/6; logic interaction 45%, held-out **0/6**): relational $\gg$ arithmetic $>$ logic at both scales. The animals row adds the decisive nuance: its interaction share (22%) is essentially arithmetic's (23%), yet its held-out transfer is perfect and arithmetic's is at chance — **what separates domains is whether the operator direction transfers, not how much interaction the state carries**. This is a **domain contrast, strictly scoped**: the factorization does not emerge for arithmetic **under our models, tokenization, prompts, and intervention procedure** — a regime where 1.7B/8B arithmetic competence is itself limited and the operation may be computed distributively or iteratively rather than retrieved. We claim the contrast, not a theorem about algorithmic computation.

Reconciling a contrary report. Christ et al. (2025) find that **add-N** operators (a fixed addend N over one number) *do* generalize. The cuts differ: add-N varies a linear numeric *parameter* — effectively an operand on the number line (Gurnee & Tegmark 2024) — while our + × - cut varies the *function*. Running both cuts on Qwen3 places add-N between relations and + × - (held-out 5/12 · 6/12), and its directions are by far the most collinear of the three families (76% of the operator-set variance on one line — a number line, not a set of operations). Their positive and our negative coexist without contradiction; the full table is in Appendix A.

4.9 Provenance note

This project began as a replication of the J-space readout claim of Gurnee et al., 2026, whose

"workspace" band we intervene on throughout; under matched controls that readout showed no advantage for our task (Appendix B). Nothing above depends on J-space.

5 Discussion

Modes of intervention, and what a margin measures. What looked like a gap between influencing a preference and executing an operation was a gap between *ways of applying the same vector*: a large added multiple moves margins while pushing the state off-manifold; a calibrated multiple, or replacement with the composed state, generates at ceiling (§4.2). The null battery likewise decomposes the margin — half uninstalling the source, half installing the target *category*, operand specificity visible only in generation (§4.1). Both points travel: margin shifts at one fixed dose measure category-level preference under a possibly off-manifold intervention, and negative generation results do not, by themselves, show the information is absent. The three-level discipline — representation, influence, sufficiency, with sufficiency tested by *composition* — is the transferable method here.

Relational computation as declension. The measurements invite a morphological reading, offered as an organizing metaphor rather than a claim: the same stem takes different endings, the same ending applies across stems (§4.5), two cases share a surface form without being the same case (**syncretism**, §4.4), stem and ending compose with a small, behaviorally inert residue of **fusion** (§4.3), and the composed stem-plus-ending is a usable form (§4.2). The limits: unlike morphological case, the marking is continuous, graded, and **layer-dependent** — a trajectory, not a static affix (§4.6) — and arithmetic and logic do not decline (§4.8). The honest formulation is that entities acquire **transient relational configurations partially comparable to grammatical case**, not that the model implements a case grammar.

6 Conclusion

An LLM represents a relational operation at three verifiable levels: the operator **exists** as a stable direction — identically the factorization's operator component; it **causally controls** the answer margin — every swap flips, localized at the query token, vanishing under permuted labels; and the factorization is **behaviorally sufficient** — a trajectory composed from its additive parts, no interac-

tion term, makes the model produce the answer at its own accuracy level, by replacement or by calibrated addition. It generalizes to unseen operands, rewordings, a second domain, and a second architecture; it does not emerge for arithmetic or logic under our setup. The gap between shifting a preference and executing an operation, where it appeared, was a property of the intervention, not the representation.

7 Limitations

Scale and family: 1.7–9B models from two decoder-only families; no ≥ 30 B model; relational linearity holds for a subset of relations even in-domain (cf. LRE $\sim 48\%$). **Sufficiency is scoped:** "generates" means short-decode containment (3 tokens) against the model's own clean accuracy as ceiling, on short single-query prompts; composition and dose $\times$ position replicate at 8B (§4.2); the leave-one-out composition reaches 36%, not the full-grid 52% — some of the composed state's precision does come from seeing the cell. **Coverage:** five country + four animal relations, English-only; three context frames and a full relexicalization (§4.5); other entity types and languages open. **Arithmetic:** single-digit regime (BPE), at scales where arithmetic competence is itself limited (§4.8). **Reading-position result** restates known subject-enrichment $\rightarrow$ attribute-extraction dynamics. **No J-space claim** (§4.9, Appendix B).

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A Reproducibility, stimuli, and implementation details

Checkpoints and hardware. Qwen/Qwen3-1.7B, Qwen/Qwen3-8B, google/gemma-2-9b, all bf16 on a single AMD Strix Halo APU (128 GB unified memory, ROCm PyTorch). Seeds default to 0 and affect only the random controls; direction building is deterministic.

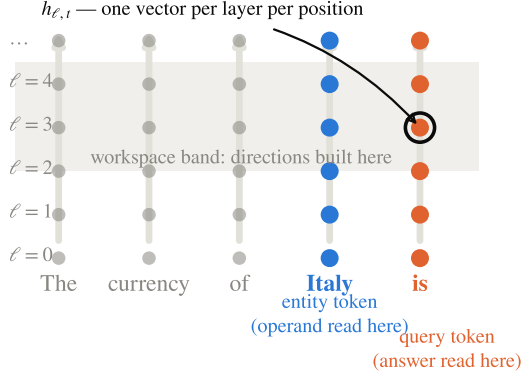
Stimuli. Operands (12 countries): Italy, Japan, Russia, China, France, Germany, Brazil, Egypt, India, Mexico, Sweden, Turkey. Relations: currency, capital, language, demonym ("word for a person from"), continent; every (country, relation) cell has a gold one-word answer (e.g. Italy → euro, Rome, Italian, Italian, Europe); the full grid ships in `data/rerelations.json`, the lexical reformulations in `data/rerelations_lex.json`, and the paraphrase frames in the dataset metadata.

Answer scoring and selection. Answers are scored at the first token of " " + answer. Within each ordered operator pair, operands whose two answers collide at that token are dropped (4 of 12 survive on the syncretic demonym↔language pairs; all 12 elsewhere). Whole-word single-token coverage: 0.92 (Qwen3 BPE), 0.95 (Gemma SentencePiece).

Directions, interventions, and controls. $v(\text{op})[\text{layer}]$ is the mean over (operand, frame) cells of the op's residual minus that cell's mean over ops, computed at ~25 evenly spaced source layers restricted to the 38–92% depth band. Interventions add $\alpha \cdot (v(\text{to}) - v(\text{from}))$ at every position of the band ($\alpha = 4$ unless swept). The matched-norm control draws one Gaussian vector per layer and rescales the band-concatenated stack to exactly the injected difference's norm. Clean mean margins $\text{logit}(\text{to}) - \text{logit}(\text{from})$ are ≈ -7 to -8 across models and frames; per-operand values for every experiment are released as long-form artifacts (`results/ablation/*_long.parquet`).

Method schematic.

The state is a grid: layers $\times$ positions



One object, applied three ways

at each band layer ℓ , average the query-token states:

$$v_{\ell}(\text{op}) = \text{mean}_o [h_{\ell, -1}(o, \text{op}) - \text{mean}_k h_{\ell, -1}(o, k)]$$

$\equiv$ the ANOVA operator main effect $b_{\ell}(\text{op})$ (verified: dev. 10^{-8})

- steer** add $\alpha (v_{\ell}(B) - v_{\ell}(A))$ at the query token $\rightarrow$ margin flips; generates at calibrated α
- compose** replace $h_{\ell, -1}$ with $\mu_{\ell} + a_{\ell}(o) + b_{\ell}(k) \rightarrow$ generates at the clean ceiling
- decode** classify / ANOVA the same states $\rightarrow$ factorization, layer profiles

Figure 2: Method in one schematic: the model's state is a token-by-layer grid of residual vectors; the entity token and the query token are the two reading positions; averaging query-token states over operands yields one operator direction per layer — identically the ANOVA operator main effect — which is then applied by addition (steering), by replacement (composition), or read out (ANOVA).

The animal-taxonomy domain. Operands (12): shark, eagle, frog, snake, bee, whale, owl, lizard, cow, salmon, penguin, deer. Relations: class, habitat, diet, covering; every cell has a stable 1-to-1 answer (e.g. shark → fish, ocean, carnivore, scales), screened so all four answers are first-token distinct per operand (single-token coverage 0.62 — multi-token answers like *carnivore* depress the greedy ceiling but not the first-token margin metric); three paraphrase frames mirror the country domain; grid in `data/animals.json`, authoring gate in `build_op_datasets.py`.

The generation audit (full table, §4.2). 1.7B relations, 224 cells/condition, $k = 8$ greedy, raw texts persisted in the artifact; forced choice = length-normalized sequence log-prob among the operand's gold answers under the same intervention:

condition	target	source	other	degraded	forced choice
clean	3.6%	58.5%	1.3%	36.6%	2.7%
patch composed (μ +operand+operator)	57.6%	11.6%	0.9%	29.9%	76.3%
patch donor	57.6%	9.8%	0.8%	31.7%	84.4%
add band $\alpha=4$ (overdose)	0.4%	0.0%	4.9%	94.6%	80.4%
add band $\alpha=0.1$ (calibrated)	59.4%	0.9%	1.3%	38.4%	87.5%
add one layer, query, $\alpha=1$	40.6%	21.9%	2.2%	35.3%	76.3%

Containment classification can also over-count (2–5% of cells contain both target and source — "the Swedish krona" contains demonym-"Swedish" as a modifier); worst-case reclassification moves target rates by ≤ 5 points.

Cross-architecture replication (full table, §4.7).

measure	Qwen3-1.7B	Qwen3-8B	Gemma-2-9B
all-pairs swap contrast	+22.6 [+14.0, +32.1]	+26.0 [+17.9, +32.8]	+30.4 [+23.8, +34.8]
flips (swap > 0)	20/20	20/20	20/20
held-out-operand contrast	+20.0 [+10.8, +29.5]	+22.8 [+14.0, +31.0]	+26.6 [+20.6, +33.2]
operator variance @ query	86%	82%	84.8%
interaction (cell residual)	9%	13%	6.8%
exponent-free marker (clean $\rightarrow$ +v)	-3.6 $\rightarrow$ +12.5	-2.9 $\rightarrow$ +8.1	-3.0 $\rightarrow$ +8.5

Table 4: Cross-architecture replication: every signature reproduces on Gemma-2-9B (logit units are not comparable across architectures; compare flip rates and variance shares).

Per-pair distributions.

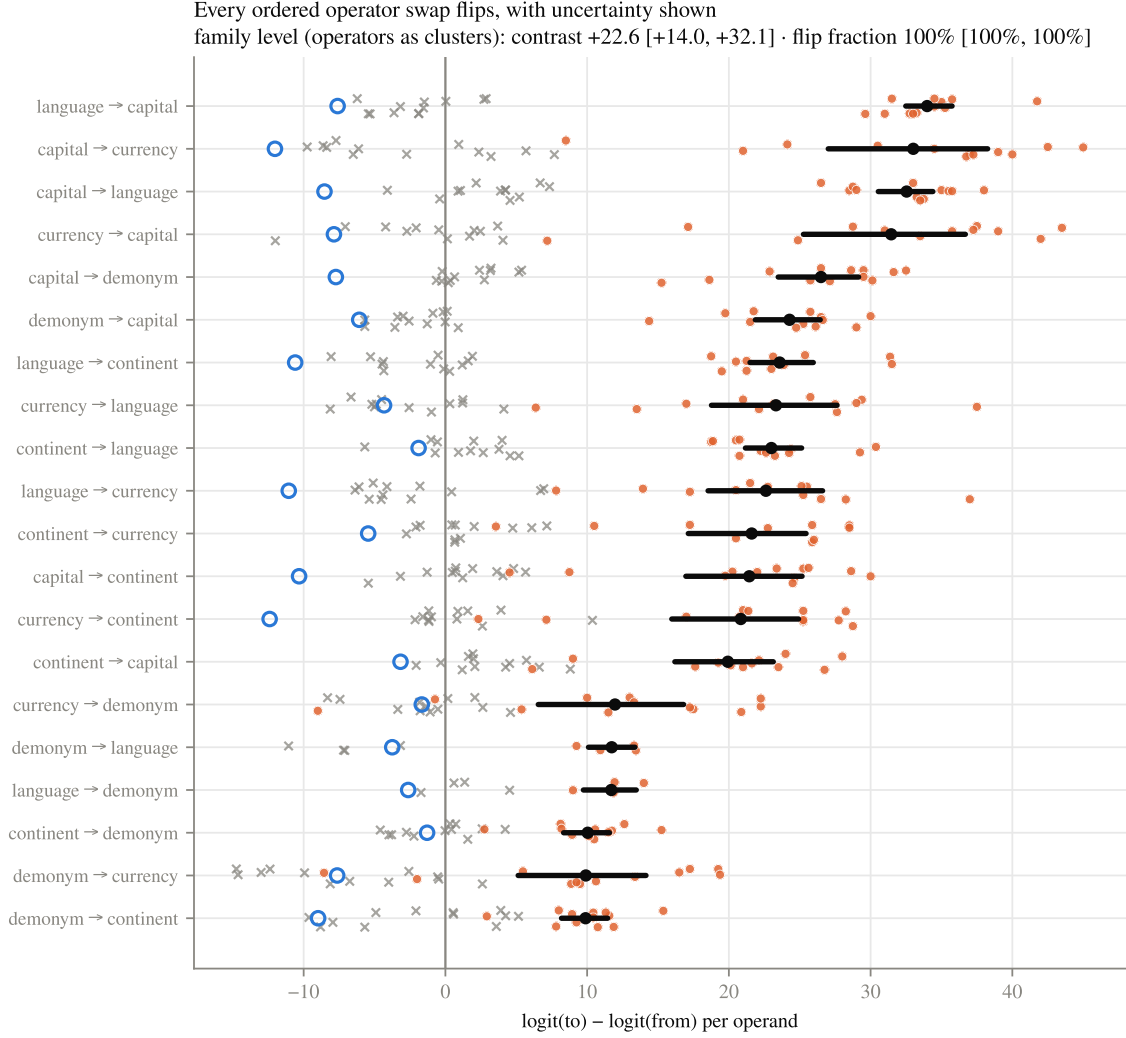


Figure 3: Per-pair distributions: swap values per operand with operand-bootstrap 95% CIs, matched-norm random control, and clean baselines. No CI crosses zero; the weakest pairs are exactly the syncretic ones (demonym $\leftrightarrow$ language and their neighbours).

The position $\times$ scope battery (§4.1; 1.7B, $\alpha = 4$). Ranks are medians, clean baseline 570; $n = 224$ (pair, operand) cells per condition:

position, scope	Δ margin [95% CI]	rank(tgt)	says target	sign+	KL off
all, band	+28.9 [+16.2, +39.9]	431	0.4%	98%	18.4
query, band	+28.7 [+16.1, +39.8]	368	2.2%	98%	0.29
query, one layer	+13.7 [+7.7, +18.3]	69	9.8%	91%	0.04
operand, band	+1.8 [+1.0, +2.5]	3194	0.4%	4%	0.26
wrong ($t = 0$), band	+2.3 [-0.2, +4.8]	194	3.6%	16%	1.56

Table 5: Where the vector must land (1.7B, $\alpha = 4$): position $\times$ layer scope, with the metrics flips hide. Ranks are medians over 224 cells; clean-baseline rank 570.

The null battery (full results, 1.7B relations, $\alpha = 4$). Contrast vs. matched-norm random with operator-level cluster-bootstrap 95% CIs; stochastic nulls pool 20 redraws:

condition	contrast [95% CI]	flip fraction
real swap	+22.62 [+14.0, +32.1]	1.00
permuted labels (per-operand)	+0.68 [-0.8, +3.9]	0.45
permuted labels (global)	+0.57 [-2.1, +3.0]	0.35
random in operator subspace	+0.75 [-2.2, +3.5]	0.45
wrong layer (early band)	+9.77 [+5.2, +13.3]	1.00
other-relation direction	+11.09 [+6.7, +17.1]	0.80
shuffled answers	+21.12 [+11.2, +32.2]	1.00

Table 6: The null battery (1.7B, $\alpha = 4$). Semantic nulls – permuted relation labels and random directions inside the operator subspace – abolish the effect; structural probes remain nonzero for mechanistically explained reasons (§4.1).

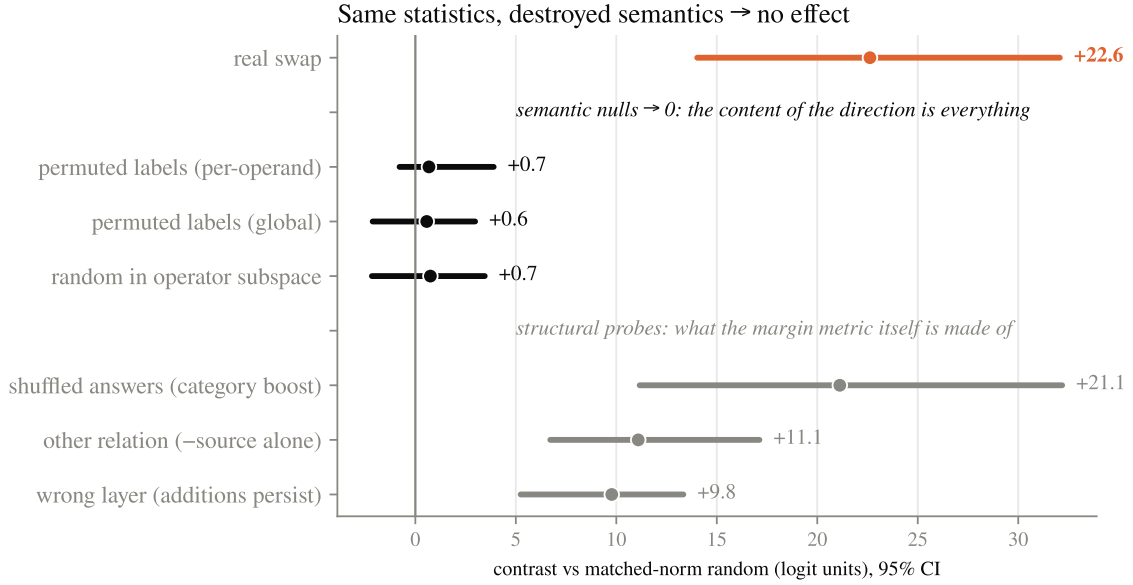


Figure 4: Forest plot of the null battery: semantic nulls (permuted labels, subspace-random) collapse to zero; structural probes remain nonzero for mechanistically explained reasons (source-uninstall component, category-level boost, additive persistence across layers).

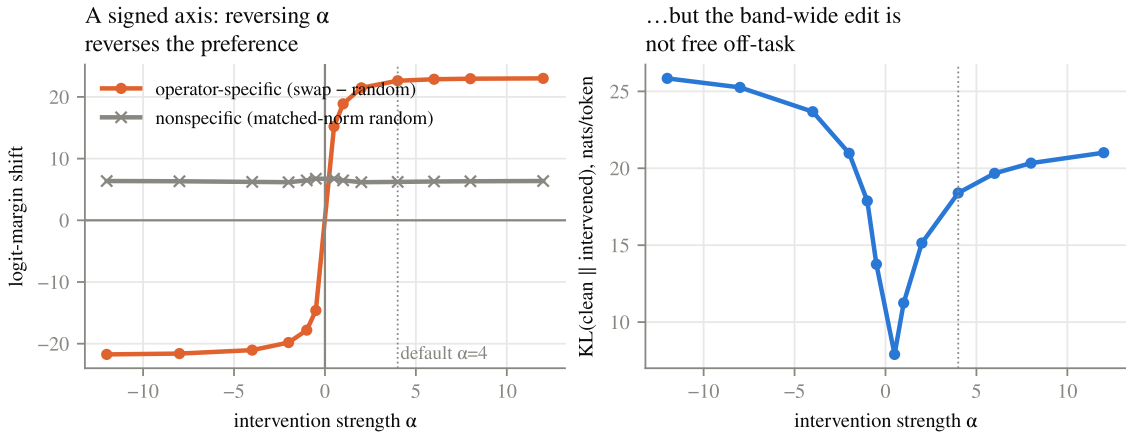


Figure 5: Dose-response with both signs: the operator-specific effect is monotone through zero and reverses with the direction's sign, saturating at both ends, while the matched-norm random control is flat; per-token KL on unrelated text grows with dose.



Figure 6: PCA of the 60 workspace vectors $H[\text{operand}, \text{operator}]$, colored by operator. At the country token the operators intermix (operand-organized, operand 59%); at the query token they separate into clean operator clusters (operator-organized, operator 86%). The faint web links each country's five operator-forms — short at the country token, splayed at the query token. Bottom: the variance split, 1.7B and 8B.

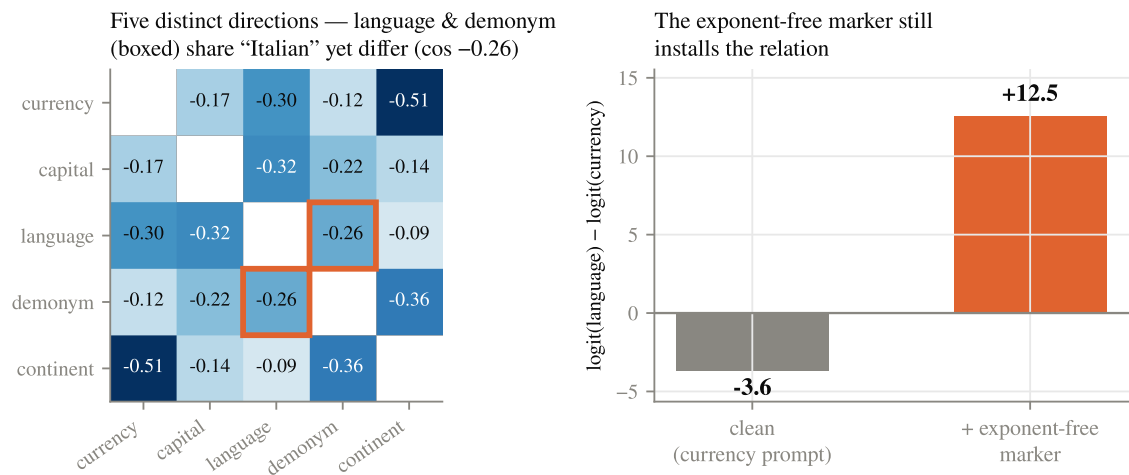


Figure 7: Left: operator-direction cosines — all five operators are distinct directions (no off-diagonal near ± 1); language and demonym, boxed, both emit "Italian" yet differ (cos -0.26). Right: the exponent-free marker, built where the two share an output word, still installs the relation causally ($-3.6 \rightarrow +12.5$).

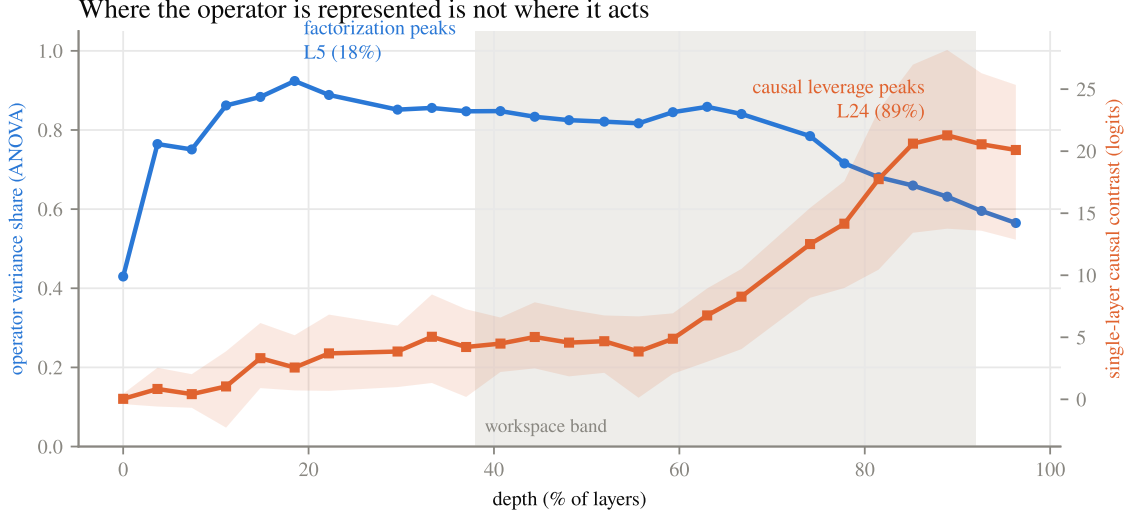


Figure 8: Layer sweep: the ANOVA operator share peaks early (92% at 19% depth) while the single-layer causal contrast peaks late (+21.3 at 89% depth); held-out-operand decodability is at ceiling at every layer.

The add-N comparison (moved from §4.8). Held-out generalization and operator-set collinearity for the three operator families:

cut	held-out generalization (1.7B / 8B)	operator-set collinearity (top-1 var)
relations (5 operations)	20/20 · 20/20	0.39 (spread paradigm)
add-N (operator = addend)	5/12 · 6/12 (+0.10 / +0.05)	0.76 (most 1-D / number-line)
+ × − (operator = function)	2/6 · 1/6 (−0.43 / −0.55)	0.64

Table 7: Reconciliation with the add-N cut: the linear-parameter family sits between relations and true two-operand arithmetic.

Qwen3 BPE restricts add-N to single-digit results (5 operands), so its generalization is weak in absolute terms; the claim is the ordering, not the magnitude.

Experiment drivers. `op_core.py` (directions $\equiv$ ANOVA components, swaps, position-restricted hooks, metric bundle, cluster bootstraps), `operator_paradigm.py`, `operator_factorize.py`, `operator_templates.py`, `operator_lexical.py`, `op_dose.py` (signed α sweep), `op_minimal.py`, `op_patch_decomp.py` (composition ladder), `op_positions.py` (position $\times$ scope $\times$ dose), `op_nulls.py` (the battery above), `op_layer_sweep.py` (decodability vs. causal effect per layer), `op_geometry_dump.py`; `verify_numbers.py` recomputes every number in this paper from the persisted artifacts.

B The J-space readout comparisons

Four readout comparisons on Qwen3-1.7B (8B where noted) asked whether the Jacobian-lens readout exposes more than the logit lens: bridge-entity pass@k (a tie, drifting toward the logit lens at 8B); a surface-form logit difference with real competitors (logit lens matches or wins in every depth band); a grammatical-number probe with Hewitt–Liang selectivity (number decodable at ceiling from *all* lenses — including a **spectrum-matched random projection**, the decisive control); and the concept-plane trajectory that motivated the project (the J-lens path is longer on random planes too — a generic magnitude effect, not a concept channel). A permuted-vocabulary lens control behaves accordingly. The 1.7B readout-geometry difference (cases reorganizing by output form in J-space) does not replicate at 8B and is not relied on. Full tables live in the project’s working log.